Electrical Power and Energy Systems 156 (2024) 109729

Contents lists available at ScienceDirect

International Journal of Electrical Power and Energy Systems

journal homepage: www.elsevier.com/locate/ijepes

Improved power loss balance control for modular multilevel converters based on variable capacitor voltage deviation predefined value under SM malfunction

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ARTICLE INFO

Keywords:
Modular multilevel converter (MMC)
Power loss balance
Submodule malfunction
Capacitor voltage deviation predefined value

ABSTRACT

When it concerns the reliable working of the modular multilevel converter (MMC), the power loss balance control cannot be ignored. Under the submodule (SM) malfunction condition, each arm of the MMC operates in an asymmetrical status, resulting in unbalanced power loss in every arm of MMCs, which affects the converter operation's reliability. This essay proposes a variable capacitor voltage deviation predefined value-based power loss balance control (VCVDPV-PLBC), which could optimize faulty arms' power losses by regulating capacitor voltage deviation predefined value to change power devices' switching loss in MMCs under SM malfunction. On this basis, the coupling relationship between SM's switching frequency and the capacitor voltage deviation predefined value is revealed. The raised VCVDPV-PLBC could effectually prolong the service life of power devices in faulty arms and strengthen the reliability of MMC system under the malfunction of the SM. The validity of the raised VCVDPV-PLBC was proved by simulation results with specialized software PSCAD/EMTDC and experimental studies with small-scale MMC setup.

1. Introduction

The Modular multilevel converter (MMC) has obtained much concentration in energy storage, new energy grid connection and so on because of its advantages of easy expansion, high power quality, lower switching frequency and convenient decoupling control of active and reactive power [1–3].

Sub-modules (SMs) are the main component of MMC, and every SM could be an underlying fault point [4]. Usually, MMC will configure about 8 % redundant SMs, and at present, the SM operation mode with the highest utilization rate and the best economy is that all non-bypass SMs take part in the system operation. When the failure ratio of SMs not higher than 8 %, under this working condition, the SMs which have breakdown are usually bypassed, which will raise the power devices' voltage strain of the faulty arm. This will cause the thermal stress of SM power devices to increase in the normal operating SMs in faulty arm, thus reducing the service life of the MMC [5–7].

Up to now, there have been some studies focusing on the power loss optimization of MMC, which are divided into hardware-based methods [8–11], methods based on modulation strategy optimization [12–15] and methods based on SM reference signal adjustment [16–20]. Reference [8] proposed a SM topology with anti-parallel low conduction voltage drop diode, which reduces the diode current in general SM and thus optimizes the power loss of the SM.

[9,10] proposed a SM topology based on wide band-gap devices, which optimizes the semiconductors' power loss in the SM according to the characteristics of wide band-gap devices including low conduction voltage drop and low switching loss. Reference [11] introduces a new modular multilevel series/parallel converter, which decreases the equivalent resistance by making the current path linked by many branches, so balance the power loss in MMC system. Literatures [12,13] proposed an interrupted modulation strategy, which diminishes the switching frequency to Reduce the number of switching times of SMs by clamping part of the SMs, thereby optimizing the MMC's loss. Reference [14] proposes a hybrid pulse width modulation that optimizes the power loss of MMC by using a fundamental frequency modulation method for part SMs to decrease the switching frequency. In response to the problem of uneven junction temperature distribution of power devices between three phases under unbalanced power grid voltage, reference [15] proposes a junction temperature balance control method for power devices in MMC. This method adjusts the switching frequency of the three phase submodules to achieve junction temperature balance between

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https://doi.org/10.1016/j.ijepes.2023.109729
Received 3 September 2023; Received in revised form 20 November 2023; Accepted 12 December 2023
Available online 22 December 2023
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three phases, thereby improving the reliability of MMC.

Reference [16] raised a power consumption balance control technology, which adjusts the inserted duty cycle of the SM by transforming the modulation to make the power consumption keep balance. Reference [17] introduces a strategy that improves the power consumption of each semiconductor in the SM by implanting the optimal SHCC into the MMC's arm current. Reference [18] introduces a method, which keeps the power loss constant by controlling the second-order harmonic cycle current in the arm, reduces the junction temperature fluctuation of the power device, and improves the lifetime of MMC. Reference [19] has proposed a loss optimization control method to address the uneven distribution of losses in the IGBT or diode within the SM. It is based on a feedforward gain controller to adjust the modulation level of the reference signal, thereby improving the loss distribution of the arm and raising the reliability of the system. Literature [20] compares the current difference integral values of power devices, and selects a more suitable mode from two bypass modes, thus proposing an optimal control mode of power loss.

Although the optimal control of power loss has achieved some achievements, the following problems still exist:

(1) Most of the existing literatures only consider the power consumption optimization for the healthy state of MMCs, ignoring the power loss optimization under the operation of SM faults as reference [21] and [22]. Under normal operation, the power loss balance control solves the power loss imbalance issue at the components level, while the power loss optimization at the arm level under SM faults is the power loss imbalance problem. So, the power loss control is not suitable for SM fault under health operation.
(2) The hardware-based way has the disadvantages of high hardware cost and complex design [20], the THD of output current will be increased by modulation strategy optimization-based method [23], for the SM reference signal adjustment-based method, when the MMC modulation is large, its effect is limited [24].

Aiming at the above limitations and deficiencies, by deducing the relationship among the SM's switching frequency, capacitor voltage deviation predefined value and power loss, this paper proposes a variable capacitor voltage deviation predefined value-based power loss balance control (VCVDPV-PLBC). The VCVDPV-PLBC can adjust capacitor voltage deviation predefined value to change power devices' switching loss in MMCs under SM malfunction, so as to optimize the power loss, and so it improves the MMC's operational reliability.

This paper's main contributions include the following five points:

(1) In this paper, by analyzing the affecting of the capacitor voltage deviation predefined value on the switching frequency of MMC, the mathematical relationship between the capacitor voltage deviation predefined value and the switching frequency is revealed under SM malfunction.
(2) This paper reveals that the bypassing of faulty SMs would causes power consumptions unbalancing among different arms, and accordingly deteriorate the reliability of the system.
(3) The VCVDPV-PLBC strategy, which can adjust the capacitor voltage deviation predefined value to regulate the switching frequency, and further regulate the power losses among arms to balance the power losses in the MMC under SM Malfunction, and improves the reliability of the system when SM malfunctions.
(4) The loss balance control method based on the capacitor voltage deviation predefined value, does not need to change the control frequency of the system, not need increase the hardware cost and design complexity of the system, and do not need to adjust the modulation strategy and control frequency of the system, which can realize low hardware cost, low system performance low system performance attenuation and efficient adjustment of the unbalanced power loss of power devices among arms in the MMC under SM Malfunction.
(5) This paper verifies the power losses unbalancing phenomenon and proposed variable capacitor voltage deviation predefined value-based power loss balance control strategy with simulation and experiment results.

This method avoids the shortcomings of research at present, and exposes the coupling relation among switching frequency, capacitor voltage deviation predefined value and the proportion of SM malfunction number, which could effectually improve the MMC's operational reliability.

This paper consists of the following parts. Section 2 briefly introduces the basic characteristics of MMC and the capacitor voltage control method of MMC adopted. Section 3 analyzes the influence of SM fault on the switching frequency of power devices and the mathematical relationship between the capacitor voltage deviation predefined value and the switching frequency. In Section 4, the power loss of MMC under SM fault is analyzed. The Section 5 of this paper puts forward the VCVDPV-PLBC. In the Section 6 and 7, the proposed strategy is proved by simulation and experiment, respectively. Section 8 shows the conclusion.

2. Basic characteristics of MMCs

2.1. Fundamental structure of MMCs

The primary topology of MMCs is demonstrated in Fig. 1(a), the three phases MMCs consist of six arms which made up of an inductor L_s and N_n SMs that have the same structure. The fundamental structure of the i-th ($i = 1, 2 \ldots\ldots N_n$) SM is illustrated in Fig. 1(b), which includes a capacitor C, two switches(T_1/T_2) and two diodes(D_1/D_2) [25].

Typically, i-th drive signal S_{aui} controls the states of the i-th SM as:

$$S_{aui} = \begin{cases} 0, T_1 \text{ is off }, \ T_2 \text{ is on} \\ 1, T_1 \text{ is on }, \ T_2 \text{ is off} \end{cases} \tag{1}$$

2.2. Capacitor voltage control method for MMCs

Fig. 2 illustrates an improved voltage balancing algorithm which could decrease the switching frequency of MMC [26]. Taking the upper arm of phase A as an reference, the number of on-state SMs N_{in} could be calculated by the reference wave y_{aur} and the modulation program. The maximum capacitor voltage deviation Δu_{max} of SM can be defined as:

$$\Delta u_{\max} = u_{\max} - u_{\min} \tag{2}$$

The capacitor voltage balance control is carried out by comparing Δu_{max} with the capacitor voltage deviation predefined value u_{set}. The u_{set} is the maximum capacitor voltage difference predefined value which is a key variable involved in the capacitor voltage balance control used in this article. In this capacitor voltage balance control, the maximum capacitor voltage difference between u_{set} and each SM in the arm at the same time will be compared Δu_{max}. If the Δu_{max} is greater than u_{set}, all SMs will be reordered by the value of capacitor voltage, and the appropriate sub-modules will be selected for input or removal according to the current direction. By adding u_{set}, the maximum voltage difference can be adjusted and the switching frequency can be controlled by changing u_{set}.

If the maximum voltage difference Δu_{max} is less than u_{set}, the ΔN_{inc} SM will be turned on or off on the basis of the last cycle. The increasing number ΔN_{inc} of the on-state SMs can be got by:

$$\Delta N_{inc} = N_{ins_new} - N_{ins_old} \tag{3}$$

According to the arm current i_{au}, the increasing number ΔN_{inc} and the u_{set}, which SM to be switched on and off could be determined as follows.

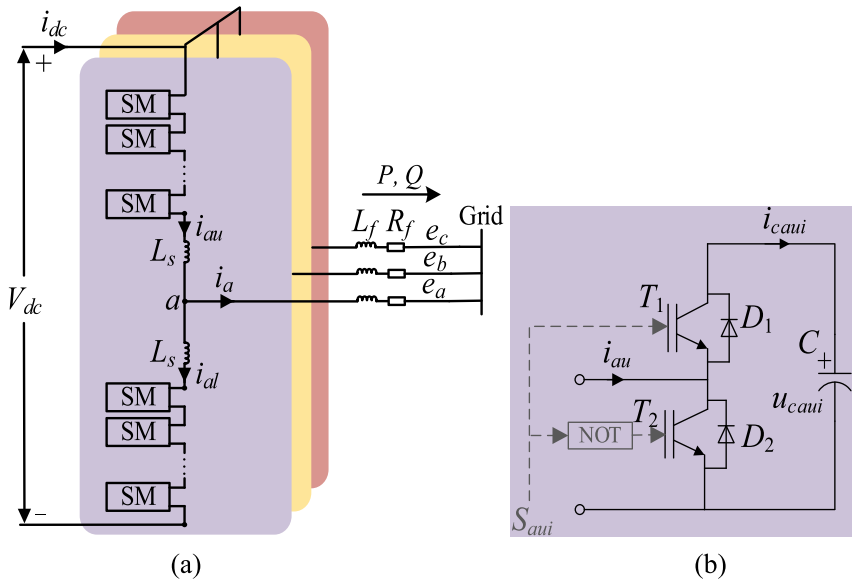


Fig. 1. (a) Basic topology. (b) SM's construction.

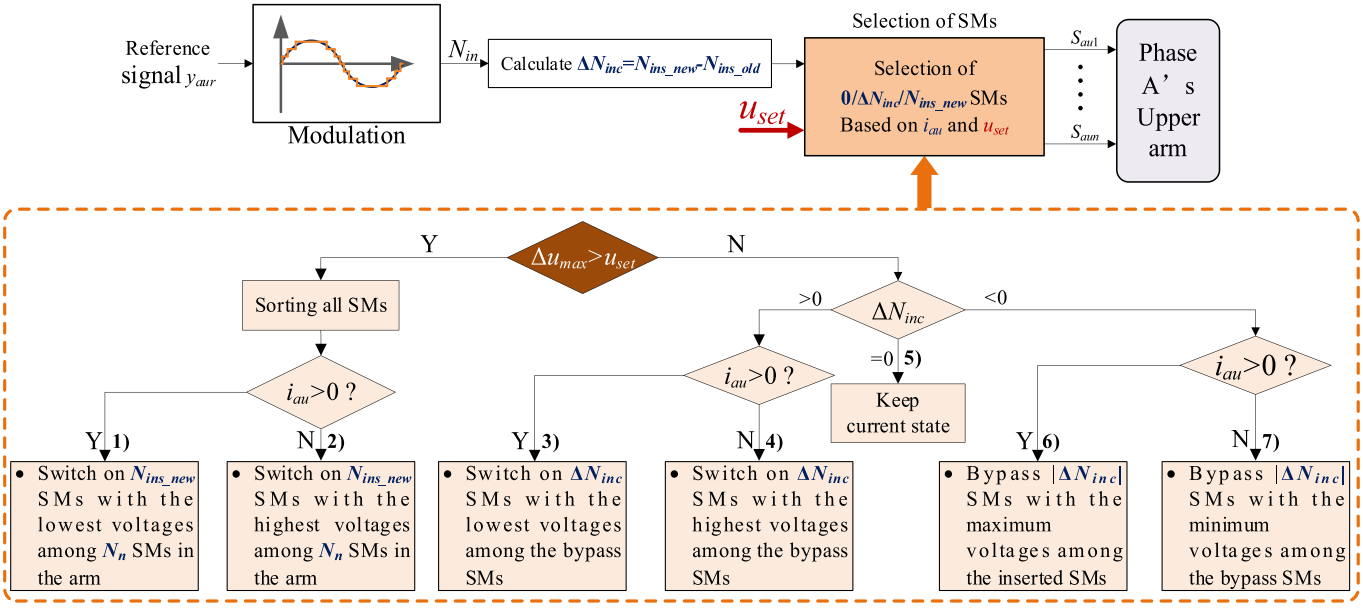


Fig. 2. Voltage-balancing control for reducing switching frequency for the MMC adopted in this manuscript.

(1) When $\Delta u_{max} > u_{set}$, all SMs are sorted according to the capacitance voltage. If $i_{au} > 0$, switch on N_{ins_new} SMs which have the lowest voltages.
(2) When $\Delta u_{max} > u_{set}$, sorting all the N_n SMs. If $i_{au} \leq 0$, switch on N_{ins_new} SMs which have the highest voltages.
(3) When $\Delta u_{max} \leq u_{set}$, if $\Delta N_{inc} > 0$ and $i_{au} > 0$, switch on ΔN_{inc} SMs which have the minimum voltages in the bypass SMs.
(4) When $\Delta u_{max} \leq u_{set}$, if $\Delta N_{inc} > 0$ and $i_{au} \leq 0$, switch on ΔN_{inc} SMs which have the maximal voltages in the bypass SMs.
(5) When $\Delta u_{max} \leq u_{set}$, if $\Delta N_{inc} = 0$, there is no SMs are put in or removed.
(6) When $\Delta u_{max} \leq u_{set}$, if $\Delta N_{inc} < 0$ and $i_{au} > 0$, bypass $|\Delta N_{inc}|$ SMs with the maximum voltages among the inserted SMs.
(7) When $\Delta u_{max} \leq u_{set}$, if $\Delta N_{inc} < 0$ and $i_{au} \leq 0$,it will bypass $|\Delta N_{inc}|$ SMs of the minimum voltages beside the inserted SMs.

2.3. Spinning reserve in the MMC

Fig. 3 demonstrates the operation of spinning reserve in case of failure used in this paper. In the spinning reserve mode, the reserve SMs are put into use as normal SMs, which will improve the utilization ratio of the SM compared with the cold reserve mode which means under cold reserve, the redundant sub-module is bypassed, and it will only be put into use when the SM involved in the arm fails. What is different from cold reserve is that, under this mode, when the MMC works under SM health status, all of the SMs, in the MMC, include redundant SMs were employed for the operation, and so that the AC and DC voltages are generated by all SMs [27]. It is just bypassing the failure SM in the failure arm, when SM faults, the submodules in other healthy arms do not need to be bypassed. Compared to conventional redundancy methods, this redundancy method keeps the average SM voltage smaller. This scheme can achieve the high SM utilization rate, high economics and low complexity of the control system in the MMC [28].

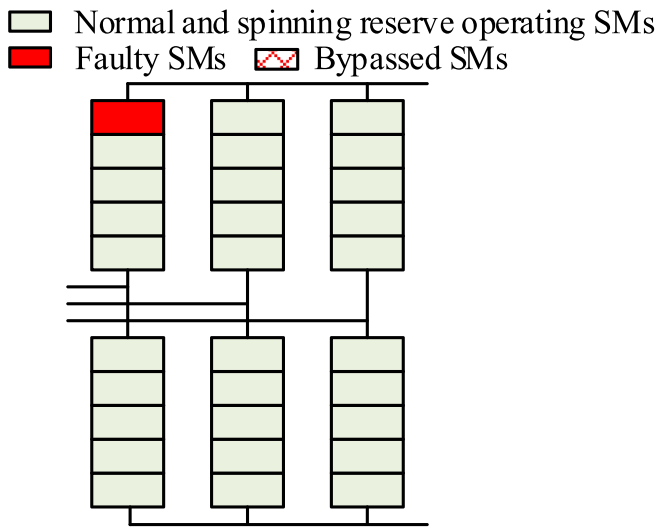


Fig. 3. Spinning reserve of MMCs.

3. Characteristics analysis of MMCs under SM faults

3.1. Analysis of SM capacitor voltage under SM faults

According to the capacitor balance control [26], hypothesis that total capacitor voltages are approximate coincidence, the rated capacitor voltage U_{cr} under normal operation can be obtained as [29]:

$$U_{cr} = V_{dc}/N_n \tag{4}$$

Taking the N_{fau} SMs fault in phase A as a reference, when SMs fault occurs, it will be bypassed by the vacuum contactor [30], and then the capacitor voltage U_f will raise compared with the normal state, and the change is related to the SM failure rate, it can be expressed as:

$$U_f = \frac{U_{cr}}{1-\lambda} \tag{5}$$

with

$$\lambda = \frac{N_{fau}}{N_n} \times 100\% \tag{6}$$

where λ represents the ratio of faulty SM number to the total number.

According to (4)~(6), the U_f will rise when the number of faulty SM increase. Fig. 4 describes the change of the capacitor voltage under various λ.

3.2. Assay of SM switching frequency under SM faults

The SM inserted into the upper arm's number could be denoted as:

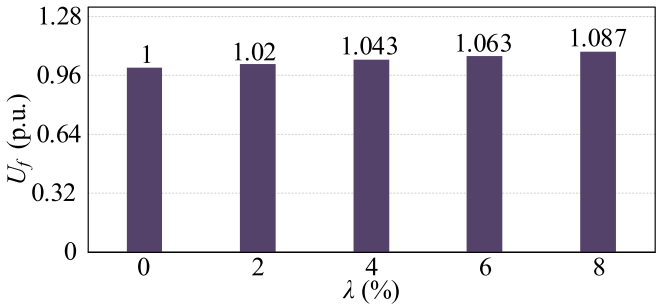


Fig. 4. SM's capacitor voltage under different λ.

$$N_{in} = round(V_{dc}\frac{1-m\cos(\omega t)}{2U_f}) \tag{7}$$

where m expresses the modulation degree, ω is expressed the radian fundamental frequency.

In the voltage balance control state, there are two forms of switching actions due to the inserted SM and bypass: one is the essential switching actions, and the other is the additional switching actions. The essential switching action indicates that the output voltage of MMC needs to increase or decrease the number of SMs, and the additional switching action is to increase or decrease the number of SMs to control the capacitor voltage balance. The essential switching can be expressed as:

$$n_b = 2round(mN_n(1-\lambda)) \tag{8}$$

Assuming that all submodules have the same parameters. If the arm current charges the SMs within a sampling period, the system will opt the N_{in} SMs which have the lowest capacitor voltage to insert, when the u_{set} exceeds a predetermined value. According to capacitor balance control, the number of additional switching actions of MMC system is:

$$n_{add} = \begin{cases} 2N_{in}round(V_{dc}\frac{1-m\cos(\omega t)}{2U_f}) \leqslant \frac{N_n}{2} \\ 2(N_n - N_{in})round(V_{dc}\frac{1-m\cos(\omega t)}{2U_f}) > \frac{N_n}{2} \end{cases} \tag{9}$$

Because the equivalent impedances of each part in the faulty arm are not equivalent, the fundamental frequency circulation will occur when the SM malfunctions. The fundamental frequency circulation, caused by SM failure, is restrained by using the strategy of [31] and using the [32] method to suppress the circulation in the second harmonic. Take phase A as an example, the i_{au} is could be obtain as:

$$i_{au} = \frac{1}{2}i_m\cos(\omega t + \varphi) + \frac{m}{4}i_m\cos\varphi \tag{10}$$

with

$$\begin{cases} i_m = \frac{\sqrt{2}\sqrt{P^2+Q^2}}{3U_s} \\ \varphi = \arctan(\frac{Q}{P}) \end{cases} \tag{11}$$

where φ expresses the grid power factor angle, i_m is the grid current amplitude, Q is the reactive power and P express the grid active power.

Fig. 5 displays the current i_{au} in one cycle T_f. According to (10) and (11), the zero crossing instants t_1 and t_2 of the i_{au} in Fig. 5 could be gained as:

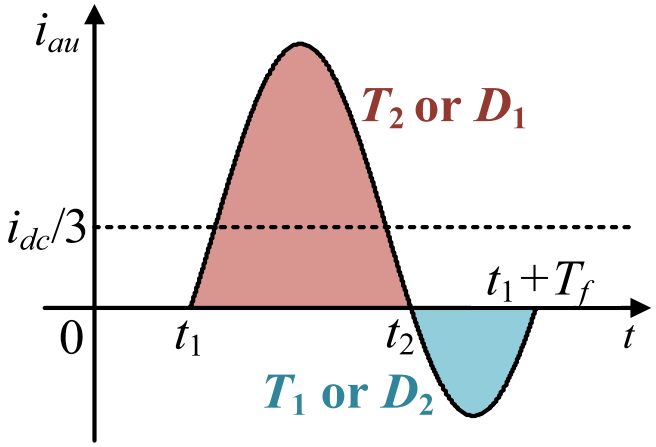


Fig. 5. MMC's upper arm current i_{au}.

$$\begin{cases} t_1 = -\dfrac{\arcsin(\dfrac{2i_{dc}}{3i_m}) + \varphi}{\omega} + T_f \\ t_2 = \dfrac{\pi + \arcsin(\dfrac{2i_{dc}}{3i_m}) - \varphi}{\omega} + T_f \end{cases} \tag{12}$$

where i_{dc} expresses the dc-link current as:

$$i_{dc} = P/V_{dc} \tag{13}$$

Assume that the current charges the capacitor, take the phase A's upper arm in MMC as a reference. By integrating the current, the change of capacitance voltage of the inserted SM in Δt time can be calculated as:

$$\Delta u = \frac{1}{C}\int_0^{\Delta t} i_{au}dt \tag{14}$$

with

$$\Delta t = \frac{C\Delta u_{\max}}{|i_{au_ave}|} \tag{15}$$

where i_{au_ave} represents the average value of the phase A's upper arm current in time Δt.

According to the equation (9) and (15), during Δt time, the additional switching frequency is related to the maximal capacitor voltage difference. According to the number of additional switches in Δt time as n_{add}, the expression of additional frequency can be obtained as follows:

$$f_{ave_add} = \frac{n_{add}}{2\Delta t} = \frac{n_{add}|i_{au_ave}|}{2C\Delta u_{\max}} \tag{16}$$

The additional switching frequency obtained according to (16) can be integrated in a period to obtain the total number of additional switching times in the period, which together with the necessary switching times in a period constitutes the total switching times in the period. The total switching times of sub-modules in a cycle can be calculated from the necessary switching times and the additional switching times, it could be approximated as:

$$n_{total_T} = \frac{1}{C\Delta u_{\max}}\int_0^{T_f} n_{add}|i_{au}|dt + n_b \tag{17}$$

According to (17), the total switching times in phase A's upper arm during a period T_f are obtained, based on the relationship between switching frequency and switching times, in the MMC system, the switching frequency could be obtained as:

$$f_{sw} = \frac{n_{total_T}}{2N_n(1-\lambda)T_f} = \frac{1}{2N_n(1-\lambda)T_f}\int_0^{T_f}\frac{n_{add}i_{au}}{C\Delta u_{\max}}dt + \frac{n_b}{2N_n(1-\lambda)T_f} \tag{18}$$

According to the zero-crossing point of the arm current i_{au} and the subsection point of n_{add}, (18) can be segmented and integrated.

This paper has made 3 fundamental assumptions as follows:

(1) Supporting that all submodule capacitor voltage is equal to V_{dc}/[N_n(1-λ].
(2) Supporting that the characteristic of every SM are equivalent.
(3) The second-order harmonic compensation component could be ignored because it is tiny.

Based on the above three assumptions and after simplification, the switching frequency of the SM could be obtained as:

$$f_{sw} = \frac{i_m}{16\pi C u_{set}}\left\{\sqrt{1-\frac{m^2\cos\varphi^2}{4}}\left(8+2m^2\cos^2\varphi\right) + m\cos\varphi\left[2\pi - 8\arccos\left(\frac{m}{2}\varphi\right) - 4m\right]\right\} + \frac{m}{T_f} \tag{19}$$

According to (19), the switching frequency is inversely proportional to u_{set}, which raises with the decrease of u_{set} and diminishes with the increase of u_{set}.

4. Power consumption assay of MMCs under SM faults

4.1. Assay of semiconductor's current

As illustrated in Fig. 5, in one period T_f, if the arm current $i_{au} > 0$, i_{au} runs through T_2 or D_1; if the arm current $i_{au} < 0$, i_{au} runs through T_1 or D_2. According to [33,34], (10)~(13), the average current I_{T1_ave}, I_{T2_ave}, I_{D1_ave}, I_{D2_ave} of the T_1, T_2, D_1, D_2 can be obtained, respectively, as:

$$\begin{cases} I_{T1_ave} = \dfrac{1}{T_f}\displaystyle\int_{t_2}^{t_1+T_f}\frac{1-y_{aur}(t)}{2}(-i_{au}(t))dt \\ I_{T2_ave} = \dfrac{1}{T_f}\displaystyle\int_{t_1}^{t_2}\frac{1+y_{aur}(t)}{2}i_{au}(t)dt \\ I_{D1_ave} = \dfrac{1}{T_f}\displaystyle\int_{t_1}^{t_2}\frac{1-y_{aur}(t)}{2}i_{au}(t)dt \\ I_{D2_ave} = \dfrac{1}{T_f}\displaystyle\int_{t_2}^{t_1+T_f}\frac{1+y_{aur}(t)}{2}(-i_{au}(t))dt \end{cases} \tag{20}$$

According to [33,34], (10)~(13), the semiconductor current I_{T1_rms}, I_{T2_rms}, I_{D1_rms}, I_{D2_rms} of the T_1, T_2, D_1, D_2 could be obtained, respectively, as follows:

$$\begin{cases} I_{T1_rms} = \sqrt{\dfrac{1}{T_f}\displaystyle\int_{t_2}^{t_1+T_f}\frac{1-y_{aur}(t)}{2}i_{au}^2(t)dt} \\ I_{T2_rms} = \sqrt{\dfrac{1}{T_f}\displaystyle\int_{t_1}^{t_2}\frac{1+y_{aur}(t)}{2}i_{au}^2(t)dt} \\ I_{D1_rms} = \sqrt{\dfrac{1}{T_f}\displaystyle\int_{t_1}^{t_2}\frac{1-y_{aur}(t)}{2}i_{au}^2(t)dt} \\ I_{D2_rms} = \sqrt{\dfrac{1}{T_f}\displaystyle\int_{t_2}^{t_1+T_f}\frac{1+y_{aur}(t)}{2}i_{au}^2(t)dt} \end{cases} \tag{21}$$

4.2. Assay of semiconductor's on-state losses

On the basis of [35], Suppose V_{d0} and V_{ce0} is on-state zero-current voltage drop of diode and IGBT. The conduction loss P_{con_T1}, P_{con_T2}, P_{con_D1}, P_{con_D2} of T_1, T_2, D_1, D_2 in the SM of the MMC can be obtained as:

$$\begin{cases} P_{con_T1/2} = V_{ce0}I_{T1/2_ave} + R_{ce0}I_{T1/2_rms}^2 \\ P_{con_D1/2} = V_{d0}I_{D1/2_ave} + R_{d0}I_{D1/2_rms}^2 \end{cases} \tag{22}$$

where R_{d0} is on-state resistance of diode, R_{ce0} is on-state resistance of IGBT.

According to (10)~(22), the conduction loss P_{con_T1}, P_{con_T2}, P_{con_D1}, P_{con_D2}, P_{con_D2} of the T_1, T_2, D_1, D_2 are almost unaffected by the SM malfunction.

4.3. Assay of semiconductor's switching losses

On the basis of [35], it is assumed that the energy required for switching the working state of IGBT is E_{sw}, and the energy required for the reverse recovery energy of diode is E_{rec}. the switching losses P_{sw_T1}, P_{sw_T2}, P_{rec_D1}, P_{rec_D2} of the T_1, T_2, D_1, D_2, respectively, could be calculated as:

$$\begin{cases} P_{sw_T1/2} = f_{sw}E_{sw}(I_{T1/2_ave}, I_{T1/2_rms})\dfrac{U_f}{U_{ref}} \\ P_{rec_D1/2} = f_{sw}E_{rec}(I_{D1/2_ave}, I_{D1/2_rms})\dfrac{U_f}{U_{ref}} \end{cases} \tag{23}$$

where U_{ref} expresses the rated operating voltage provided by manufacturer.

According to (23), it could be seen that the switching loss have a bearing on the f_{sw} and U_f. An increase in U_f or f_{sw} leads to an increase switching losses in SM, and a decrease in U_f or f_{sw} leads to a decrease in SM switching losses.

4.4. Assay of semiconductor's total power consumption

The total power consumption P_{T1}, P_{T2}, P_{D1}, P_{D2} of the T_1, T_2, D_1, D_2 can be obtained as:

$$\begin{cases} P_{T1/2} = P_{con_T1/2} + P_{sw_T1/2} \\ P_{D1/2} = P_{con_D1/2} + P_{rec_D1/2} \end{cases} \tag{24}$$

According to (24), there is almost no influence on the conduction loss under the SM fault condition, but the switching loss will ascend with the augment of failures SMs, thus affecting the increasing of the total loss generated by the diode and the IGBT, as illustrated in the following Table 1.

When λ ascends, the switch power consumption $P_{sw_T1/2}$ and $P_{rec_D1/2}$ increasing accordingly, and at the same time, the on-state loss $P_{con_T1/2}$ and $P_{con_D1/2}$ are almost unaffected. Therefore, the total power consumption $P_{T1/2}$ and $P_{D1/2}$ of the semiconductors could be augmented with the augmentation of λ.

When λ drops, the switch power consumption $P_{sw_T1/2}$ and $P_{rec_D1/2}$ decreasing accordingly, and at the same time, the on-state loss $P_{con_T1/2}$ and $P_{con_D1/2}$ are almost unaffected. Therefore, the total power consumption $P_{T1/2}$ and $P_{D1/2}$ of the semiconductors would be descended with the drop of λ.

On the basis of (24), the power loss of the T_1, T_2, D_1, D_2 under different λ as demonstrated in Fig. 6. Obviously, the power loss of the T_1, T_2, D_1, D_2 ascends with the augmentation of λ, and the total power loss of the T_1, T_2, D_1, D_2 drops along with the decreasing of λ, which could be clearly seen in Table 1.

4.5. Assay of arm energy

The energy of capacitors in arm under normal operation could be expressed as:

$$W_{rated} = \frac{1}{2} C N_n U_{cr}^2 \tag{25}$$

Taking the N_{fau} SMs fault in phase A as the example, when SMs fault occurs, it will be bypassed. The total energy of the capacitors under SM faults operation is defined as:

$$W_{au} = \frac{1}{2} C (N_n - N_{fau}) \left(\frac{U_{cr}}{1-\lambda} \right)^2 \tag{26}$$

According to (25)~(26), the W_{au} will change when the number of faulty SM change. Fig. 7 describes the change of the arm energy under various λ. It could be concluded that the unbalanced energy distribution of different arms is related to the number of sub-module failures in arms. The analysis results of the energy balance with other phases will also follow similar pattern.

Table 1
Relationship between $P_{T1/2}$, $P_{D1/2}$ and λ.

λ	$P_{sw_T1/2}$	$P_{rec_D1/2}$	$P_{con_T1/2}$	$P_{con_D1/2}$	$P_{T1/2}$	$P_{D1/2}$
↑	↑	↑	—	—	↑	↑
↓	↓	↓	—	—	↓	↓

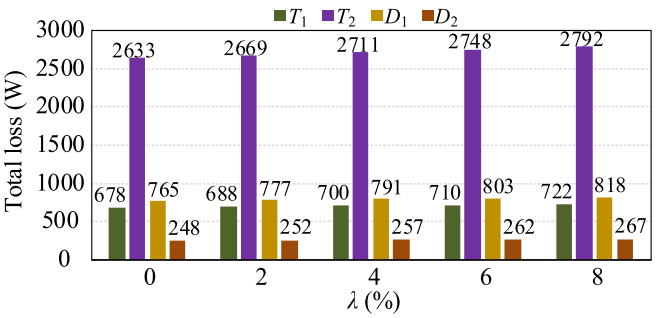


Fig. 6. Total loss of the devices T_1, T_2, D_1, D_2 in the SM under different λ.

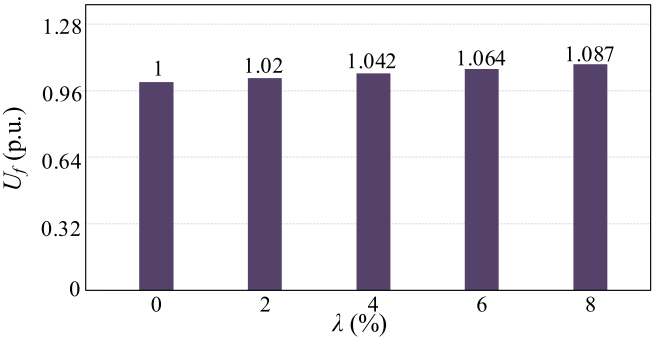


Fig. 7. Arm energy under different λ.

5. Proposed VCVDPV-PLBC strategy for MMCs under SM malfunction

5.1. Proposed VCVDPV-PLBC approach

According to (19) and the simulation research in Section 6, the f_{sw} under different λ and u_{set} are shown in Fig. 8. The f_{sw} rises with the reduction of u_{set}, and the f_{sw} drops with the ascend of u_{set}.

On account of the relationship among various λ, u_{set} and switching frequency, a VCVDPV-PLBC approach is proposed to adjusts the power loss $P_{T1/2}$ and $P_{D1/2}$ of the power device in MMC by using the predefined value of capacitor voltage deviation, as demonstrated in Table 2.

(1) Climb of u_{set}: the switch power consumption $P_{sw_T1/2}$ and $P_{rec_D1/2}$ drop, and at the same time, the on-state loss $P_{con_T1/2}$ and $P_{con_D1/2}$ are almost unaffected. Therefore, the semiconductor's power consumption $P_{T1/2}$ and $P_{D1/2}$ drop.
(2) Drop of u_{set}: the switch power consumption $P_{sw_T1/2}$ and $P_{rec_D1/2}$ ascend, and at the same time, the on-state loss $P_{con_T1/2}$ and

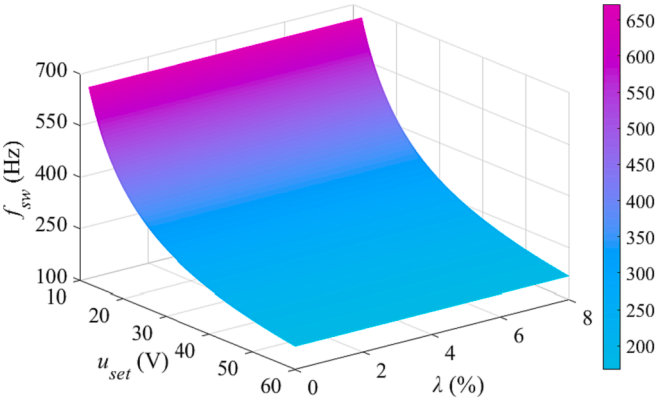


Fig. 8. Switching frequency with different λ and u_{set}.

Table 2
Relationship between $P_{T1/2}$, $P_{D1/2}$ and u_{set}.

u_{set}	$P_{sw_T1/2}$	$P_{rec_D1/2}$	$P_{con_T1/2}$	$P_{con_D1/2}$	$P_{T1/2}$	$P_{D1/2}$
↑	↓	↓	—	—	↓	↓
↓	↑	↑	—	—	↑	↑

$P_{con_D1/2}$ are almost unaffected. Therefore, the semiconductor's power consumption $P_{T1/2}$ and $P_{D1/2}$ ascend.

5.2. Raised VCVDPV-PLBC tactics for MMCs under SM malfunction

According to the above analysis, this paper proposes a VCVDPV-PLBC strategy to optimize power consumption spread under SM malfunction, as demonstrated in Fig. 9.

The greatest power consumption of the semiconductors is defined as $P_{SM} = \text{Max}\{P_{T1}, P_{T2}, P_{D1}, P_{D2}\}$, where the power loss P_{T1}, P_{T2}, P_{D1} and P_{D2} could be obtained by (24). The power consumption of the SMs is symmetrical, and so take the power loss of a SM in every arm to considerate. The reference power consumption can be obtained according to the power loss of each arm in the phase A, B and C of MMC system. The reference power consumption P_{l_ref} could be obtained as:

$$P_{l_ref} = \text{Min}[P_{l_au}, P_{l_al}, P_{l_bu}, P_{l_bl}, P_{l_cu}, P_{l_cl}] \tag{27}$$

where P_{l_au}, P_{l_bu} and P_{l_cu} are three-phase upper arm losses, P_{l_al}, P_{l_bl} and P_{l_cl} are three-phase lower arm losses, respectively.

The PI controller could regulate its u_{set}, for the MMC system in the faulty arm, as demonstrated in Table 2, to realize the power losses balancing based on VCVDPV-PLBC approach. The u_{setr} is the rated capacitor voltage deviation predefined value of the VCVDPV-PLBC for the MMC. A compensation part u_{setjk} ($j = a, b, c$; $k = u, l$) of theu_{set} generated by the PI controller. If the P_{l_jk} is larger than the P_{l_ref}, the PI controller would raise the u_{setjk}, and the capacitor voltage deviation predefined value $u_{sets_jk} = u_{setr} + u_{setjk}$ is increased, the P_{l_jk} is decreased to follow P_{l_ref}. It could be observed that the VCVDPV-PLBC can realize power consumption balancing effectively among the arms of the MMC when SM malfunction.

5.3. Lifetime analysis of proposed VCVDPV-PLBC

On the basis of [36], the average PN junction heat $T_{jm_T/D}$ and the PN junction heat fluctuation $\Delta T_{j_T/D}$ switch/diode could be obtained as:

$$T_{jm_T/D} = P_{T/D} \cdot \sum_{i=1}^{4} R_{tc_T/D}(i) + P_{T/D} \cdot R_{thch} + T_h \tag{28}$$

$$\Delta T_{j_T/D} = 2P_{T/D} \cdot \frac{\left(1 - e^{-\frac{t_{on_T/D}}{\tau_{thjc_T/D}(i)}}\right)^2}{1 - e^{-\frac{T_f}{\tau_{thjc_T/D}(i)}}} \cdot \sum_{i=1}^{4} R_{tc_T/D}(i) \tag{29}$$

where $t_{on_T/D}$ expresses the on-state time of the switch or diode in one period T_f. R_{thch} is the resistance of thermal from substrate to fin, $R_{tc_T/D}$ is the thermal resistance from the PN junction to substrate, $\tau_{thjc_T/D}$ express the thermal time constant of the semiconductor and T_h expresses the radiator temperature.

According to [37], (28) and (29), The number of failure cycles $N_{f_T/D}$ of the switch/diode could be obtained as:

$$N_{f_T/D} = A \cdot (\Delta T_{j_T/D})^{\alpha} \cdot e^{\left(\frac{E_a}{k_b \cdot T_{jm}}\right)} \tag{30}$$

where k_b is Boltzmann constant and E_a is activation energy. α and A are available from the manufacturer's power device test data sheet.

When the MMC works at inverter state, the T_2 take the shortest lifetime, and the semiconductor with the shortest life determines the lower limit of the SM's lifetime. Fig. 10 illustrates the T_2' lifetime of the SM without and with VCVDPV-PLBC, respectively. Obviously, the lifetime of the T_2 with proposed VCVDPV-PLBC can be improved under different λ.

5.4. Comparison with some existing methods

We have evaluated the comprehensive performance of the loss control degrees of freedom introduced in the introduction, as shown in the Table 3. The measurement indicators include three aspects: economy, implement ability, and compatibility. Economy refers to whether hardware costs need to be increased during the degree of freedom adjustment process, implement ability refers to the engineering implementation difficulty of degree of freedom adjustment, and compatibility refers to whether the degree of freedom adjustment process will have an

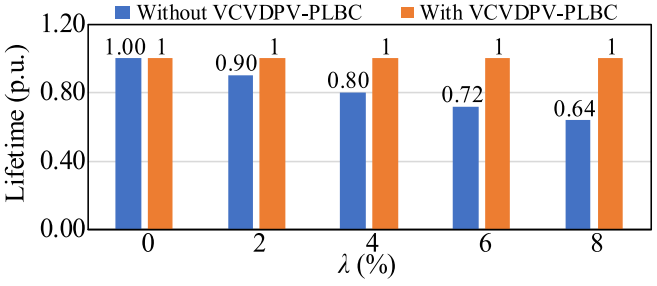


Fig. 10. SM's lifetime without and with VCVDPV-PLBC.

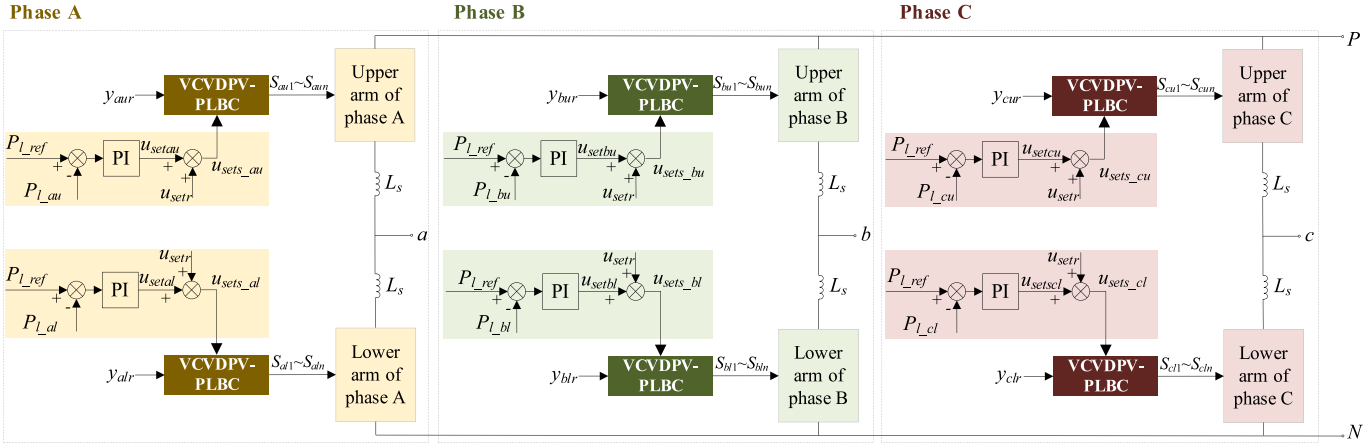


Fig. 9. The schematic design of the proposed VCVDPV-PLBC strategy.

Table 3
Method comparison.

Method	Economy	Realizability	Compatibility
Hardware-based methods [8–11]	Low	Low	Medium
Modulation strategy optimization methods [12–15]	Medium	Medium	Low
SM reference signal adjustment methods [16–20]	Medium	Medium	Low
Proposed method	High	High	High

impact on other performance of the converter. In the Table 3, Low, Medium, and High are the evaluation indicators, where High representing the best competence of the power loss control degree of freedom and Low representing the worst competence of the power loss control degree of freedom.

Hardware based methods directly increase the cost of the system, making them low cost-effective. The optimization of modulation strategy and the adjustment of reference value submodules will increase the fluctuation of capacitor voltage, increase the demand for capacitor capacitance value, and indirectly increase the system. Therefore, the economy is medium. The method proposed in this article has minimal impact on capacitor voltage fluctuations, therefore it has the high economic efficiency.

The hardware-based method requires additional hardware selection, electromagnetic compatibility analysis, driver circuit design, and other processes, which will increase the complexity of the design and therefore have low realizability. The methods for optimizing modulation strategies and adjusting reference value submodules require real-time calculations in the software to determine modulation strategies and virtual capacitor voltages at different times, which increases software complexity and computational burden. Therefore, the realizability of these two types of methods is medium. The method proposed in this paper only needs to import the relationship between the capacitor voltage deviation predefined value and switching frequency into the software program during implementation, and can adjust the loss in real time. Therefore, it has the high feasibility.

Hardware based methods can lead to inconsistent impedance of the current flow path in the SMs, resulting in a certain degree of distortion in the output voltage waveform, thus achieving medium compatibility. The methods of modulation strategy optimization and reference value SM adjustment will increase the THD output of MMC, so these two types of methods have the low compatibility. The proposed method acts at the software level, without the need for hardware circuit changes, and has minimal negative impact on the output current THD, making it the high compatibility.

6. Simulation studies

The MMC simulation model is founded by using the specialized software PSCAD/EMTDC, as demonstrated in Fig. 11, to put to the proof for the availability of the raised VCVDPV-PLBC. Table 4 shows the parameters of the simulation system. The power loss of semiconductor devices in SMs could be estimated based on (24) and the power device parameters from the manufacturer's technical manual.

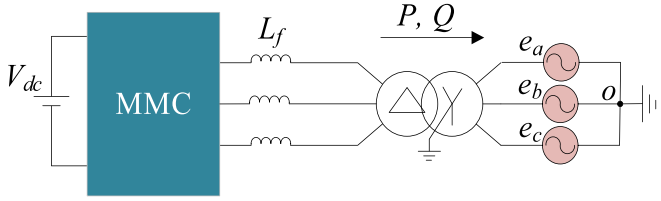


Fig. 11. Schematic design of the simulation system.

Table 4
Parameters in simulation of MMC.

Arguments	Numerical Value
Transmission power of MMC (MW)	100
DC-link's Voltage V_{dc} (kV)	150
Line voltage of the power grid (kV)	75
Per arm's SMs number N_n	50
Redundant SMs' rated number N_{fN}	5
Nominal SM capacitance C (mF)	3
Inductance L_s (mH)	45
Inductance L_f (mH)	10
Semiconductor device	FZ600R65KF2

6.1. Case I: Healthy operation situation

The simulation results of the MMC system under health conditions is illustrated in Fig. 12 where the MMC operates at 100 MW active power. The results of grid side current i_a, i_b, i_c are demonstrated in Fig. 12(a). The capacitor voltages $u_{cau1} \sim u_{cau50}$ and $u_{cal1} \sim u_{cal50}$ are demonstrated in Fig. 12(b). Fig. 12(c) illustrates arm current i_{au} and i_{al} in phase A. Fig. 12(d) illustrates the A-phase circulation i_{c_a}, B-phase circulation i_{c_b} and C-phase circulation i_{c_c} of the MMC. Fig. 12(e) describes the MMC's three-phase circulation in Zoom1. Fig. 12(f) demonstrates the power loss P_{l_au} and P_{l_al} in phase A. From the research result in Fig. 12(f), obviously, the P_{l_au} and P_{l_al} are almost identical.

6.2. Case II: 8 % malfunction in the upper arm of phase a

Fig. 13 illustrates the research results when the MMC simulation system has 8 % SM malfunction. In this case, 8 % SM malfunction in phase A, and the failures occur in $SM_{47} \sim SM_{50}$. Since 2 s, the proposed VCVDPV-PLBC is enabled. After 0.5 s, it reaches a steady state. Fig. 13(a) illustrates the capacitor voltage $u_{cau1} \sim u_{cau46}$ and $u_{cal1} \sim u_{cal50}$ of the SM under the faulty condition between 1 s and 4 s, obviously, because of the decrease of the number of SMs in the faulty arm, the capacitor voltage increases. Fig. 13(b) describes the dynamic process after the VCVDPV-PLBC is enabled in Zoom1 from Fig. 13(a). Fig. 13(c) demonstrates the waveforms after reaching the steady state in and Zoom2 from Fig. 13(a). In Fig. 13(d), the arm current i_{au} and i_{al} between 1 s and 4 s are demonstrated. Fig. 13(e) describes the waveform when the arm current reaches a steady state in Zoom1 from Fig. 13(d). Fig. 13(f) describes the three-phase circulation of the MMC under fault conditions. Fig. 13(g) describes the dynamic behaviour of the circulation within 1.8–2.2 s in Zoom1 of Fig. 13(f). Fig. 13(h) shows the steady-state waveform of the MMC's three-phase circulation in Zoom2 of Fig. 13(f). As can be seen from Fig. 13 (f) ~ (g), the proposed VCVDPV-PLBC has few influences on the circulation currents. Fig. 13(i) describes the grid-side current of the three phases in the MMC. Fig. 13(j) describes the dynamic behaviour of grid current in Zoom1 from Fig. 13(i). Fig. 13(k) shows the waveform in Zoom2 from Fig. 13(i) under the grid current reaches steady state. Fig. 13(l) illustrates the u_{set_au} and u_{set_al} in-phase A. Due to SM malfunction, P_{l_au} and P_{l_al} were unbalanced before 2 s, and the maximum loss difference was 0.16 kW. From 2 s, due to the adoption of VCVDPV-PLBC, the power consumption P_{l_au} and P_{l_al} go to balance within 0.5 s, as shown in Fig. 13(m). Since 2 s, the proposed VCVDPV-PLBC method works to reach the power loss balance between the faulty arm and the healthy arm, thus, the u_{set_au} in Fig. 13(k) is gradually increased, which proves the effectiveness of the VCVDPV-PLBC.

6.3. Case III: 8 % SM malfunction in the upper and 6 % SM malfunction in the lower arm of phase A

Fig. 14 describes the simulation results of the MMC when 8 % SM of the upper arm and 6 % SM of the lower arm malfunction in phase A. In this case, the upper arm SM47 ~ 50 of phase A malfunction, and the lower arm SM48 ~ 50 malfunction, and the other phases are healthy.

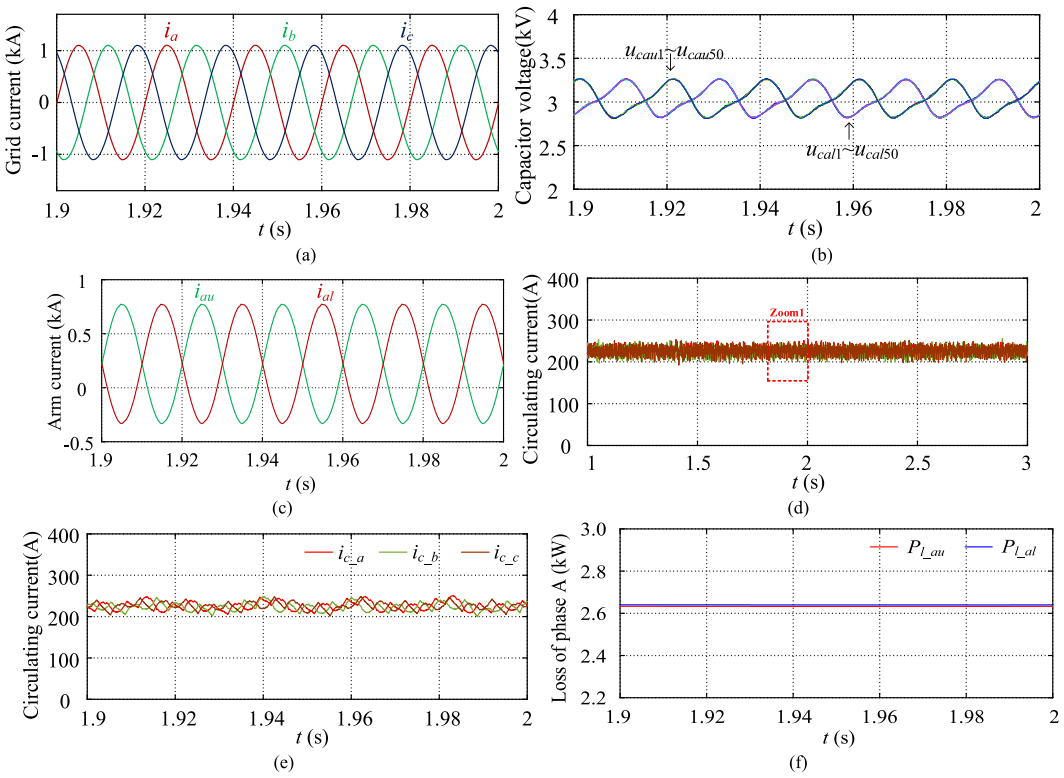


Fig. 12. (a) Grid currents. (b) SM's capacitor voltages. (c) Arm current. (d) Circulating current. (e) Circulating current in Zoom1. (f) Power loss in phase A.

The strategy VCVDPV-PLBC starts to be enabled in 2 s and achieve a steady state after 0.5 s. Fig. 14(a) shows the capacitor voltage of the SM. Fig. 14(b) demonstrates the dynamic behaviour of the capacitor voltage of the SM after the VCVDPV-PLBC strategy is enabled in Zoom1 from Fig. 14(a). Fig. 14(c) illustrates the capacitor voltage reaches steady state in Zoom2 from Fig. 14(a). Fig. 14(d) describes the grid-side current in 3-phase of the MMC. Fig. 14(e) illustrates the dynamic behaviour of grid current in Zoom1 from Fig. 14(d). Fig. 14(f) illustrates the waveform in Zoom2 from Fig. 14(d) under the grid current reaches steady state. Fig. 14(g) shows the arm currents of the upper and lower arm of phase A and the upper arm of phase B, respectively. Fig. 14(h) illustrates the dynamic behaviour of the arm current in Zoom1 from Fig. 14(g). Fig. 14(i) shows the waveform under the arm current reaches steady state in Zoom2 from Fig. 14(g). Fig. 14(j) shows the dynamic behaviour of the power loss, and the upper arm in phase B serves as a reference value of the loss of the healthy arm. The power loss of the upper arm in phase A rises to 2.792 kW and the lower arm rises to 2.748 kW due to the failure of the SM from Fig. 14(j). The proposed VCVDPV-PLBC strategy is enabled at 2 s, and after 0.5 s, the loss of the upper and lower arm of phase A is balanced to be the same as the upper arm in phase B. Fig. 14(k) describes the dynamic behaviour of the u_{set} of the upper and lower arm of phase A. Obviously, the u_{set} gradually rises after the proposed VCVDPV-PLBC strategy is enabled, and reaches steady state after 0.5 s, and is stable at 17.48 and 17.07, respectively.

6.4. Case IV: SM malfunction under power change

Fig. 15 describes the simulation results under MMC power changes. At 1.5 s, the power is adjusted from 1.0p.u to 0.8p.u, and the proposed VCVDPV-PLBC strategy is enabled at 2 s and then reaches steady state. Fig. 15(a) illustrates the SM capacitor voltage. Fig. 15(b) demonstrates the dynamic behavior of the capacitor voltage of the SMs after the proposed VCVDPV-PLBC strategy is enabled in Zoom1 from Fig. 15(a). Fig. 15(c) describes the capacitor voltage under steady state in Zoom2 of Fig. 15(a). Fig. 15(d) describes the grid-side current in 3-phase of the MMC. Fig. 15(e) describes the dynamic behavior of grid current in Zoom1 of Fig. 15(d). Fig. 15(f) shows the waveform in Zoom2 of Fig. 15(d) under the grid current reaches steady state. Fig. 15(g) describes the arm current in phase A. Fig. 15(h) illustrates the dynamic behavior of the arm current in Zoom1 of Fig. 15(g). Fig. 15(i) describes the waveform under the arm current reaches steady state in Zoom2 of Fig. 15(g). Fig. 15(j) describes the dynamic behavior of power loss in phase A. The proposed VCVDPV-PLBC strategy is enabled between 1 s and 1.5 s, therefore the power loss of P_{l_au} and P_{l_au} is balanced. The proposed VCVDPV-PLBC strategy is not enabled between 1.5 s and 2 s, therefore the power loss of P_{l_au} and P_{l_au} become unbalanced. The proposed VCVDPV-PLBC strategy is enabled at 2 s, and after 0.5 s, the power loss of the upper arm of phase A is balanced to the same as the lower arm of phase A. Fig. 15(k) describes the dynamic behavior of u_{set} in phase A. Obviously, the u_{set} in phase A gradually rises after the proposed VCVDPV-PLBC strategy is enabled, and reaches steady state after 0.5 s, and ultimately stabilizes at 18.06.

6.5. THD analysis of MMCs

Fig. 16 depicts the THD of the output current i_a of the MMC at various switching frequencies between 470 Hz and 520 Hz. From Fig. 16, obviously, the VCVDPV-PLBC has few influences on the AC output current.

7. Experiment studies

Fig. 17 shows the physical diagram of the MMC used in the experimental facility, with its output connected to a resistive load. The SM uses the IGBT module FF75R12YT30 as the power device. The driving signal of SM from the digital signal processing controller is transmitted to the gate driver through drive signal line. Table 5 gives the specific experimental parameters. The power loss of the semiconductor devices in the experiment are got through manufacturer's manual data, experimental measurements and equations (24), and the accuracy of the calculated

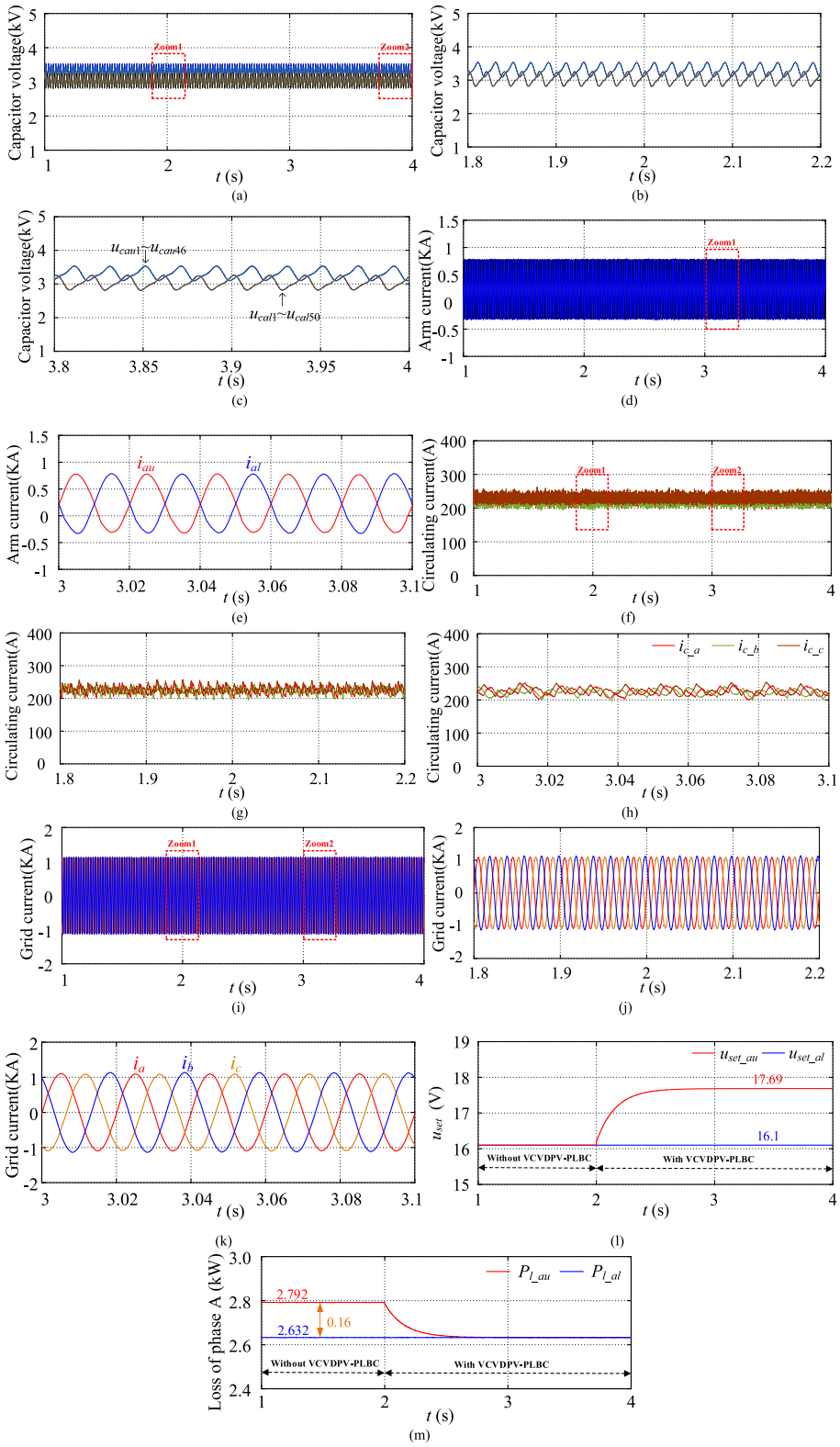


Fig. 13. (a) SM's capacitor voltages. (b) SM's capacitor voltages in Zoom1. (c) SM's capacitor voltages in Zoom2. (d) Arm current. (e) Arm current in Zoom1. (f) Circulating current (g) Circulating current in Zoom1 (h) Circulating current in Zoom2 (i) Grid currents. (j) Grid currents in Zoom1. (k) Grid currents in Zoom2. (l) Predefined value of capacitor voltage deviation. (m) Power loss in phase A.

data ensured by averaging the experimental results for many times.

7.1. Case I: Healthy condition

Fig. 18 describes the experimental results of the MMC system under normal conditions. Fig. 18(a) illustrates the output voltage u_{cau}, upper

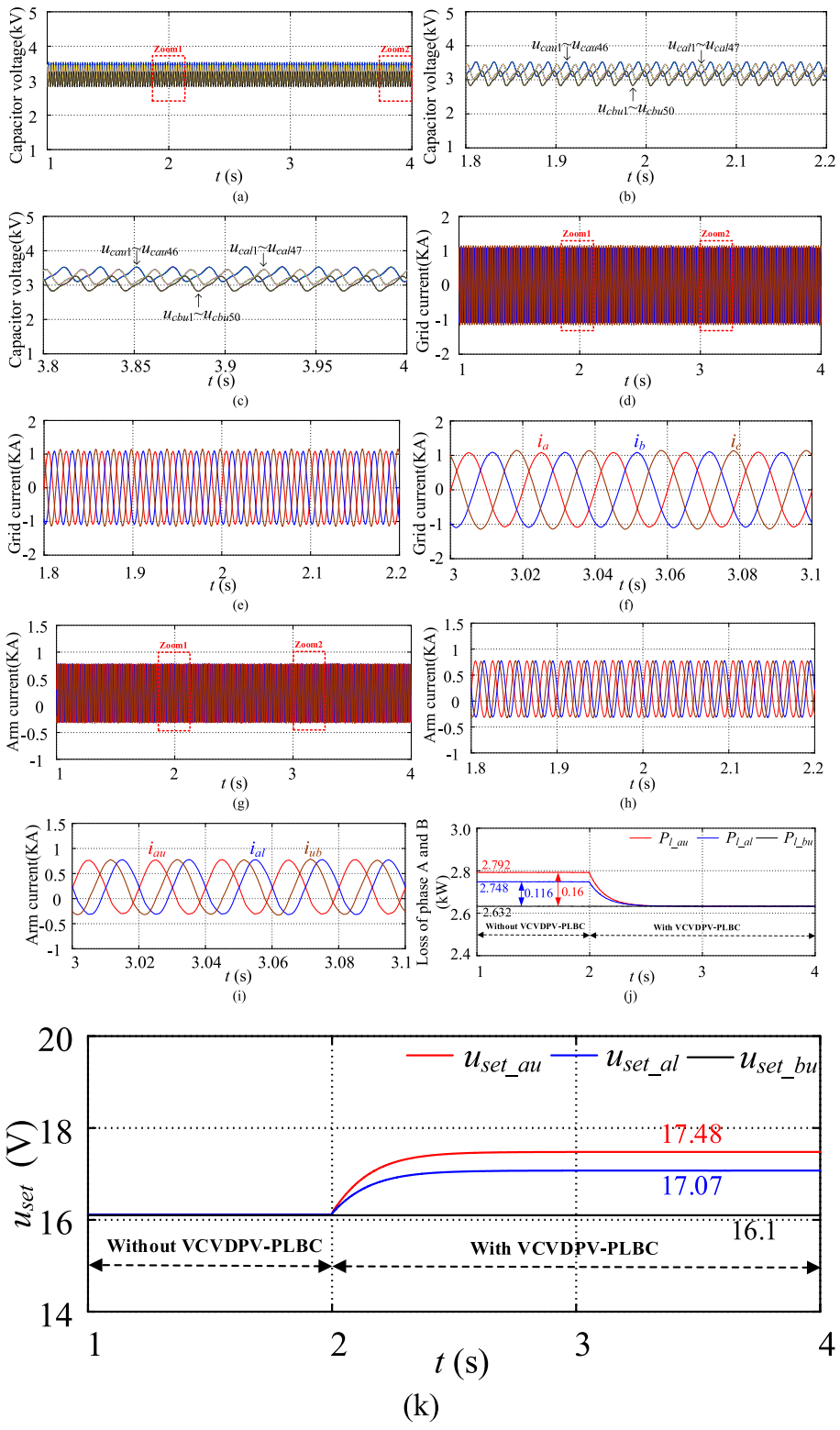


Fig. 14. (a) SM's capacitor voltages. (b) SM's capacitor voltages in Zoom1. (c) SM's capacitor voltages in Zoom2. (d) Grid currents. (e) Grid currents in Zoom1. (f) Grid currents in Zoom2. (g) Arm current. (h) Arm current in Zoom1. (i) Arm current in Zoom2. (j) Power loss. (k) Predefined value of capacitor voltage deviation.

current i_{au} and lower arm current i_{al}, SM capacitor voltage $u_{cau1} \sim u_{cau3}$, output current i_a. Fig. 18(b) illustrates the power loss P_{l_au} and P_{l_al} in the arm of the MMC, it can be seen that in this situation, they are almost the same.

7.2. Case II: SM faults situation with dynamic waveform

Fig. 19 illustrates the running state of MMC under SM malfunction condition with dynamic waveform under proposed VCVDPV-PLBC. Fig. 19(a) demonstrates the output voltage u_{cau}, the arm current i_{au} and i_{al}, the SM capacitor voltage $u_{cau1} \sim u_{cau2}$ and u_{cal1}, the output

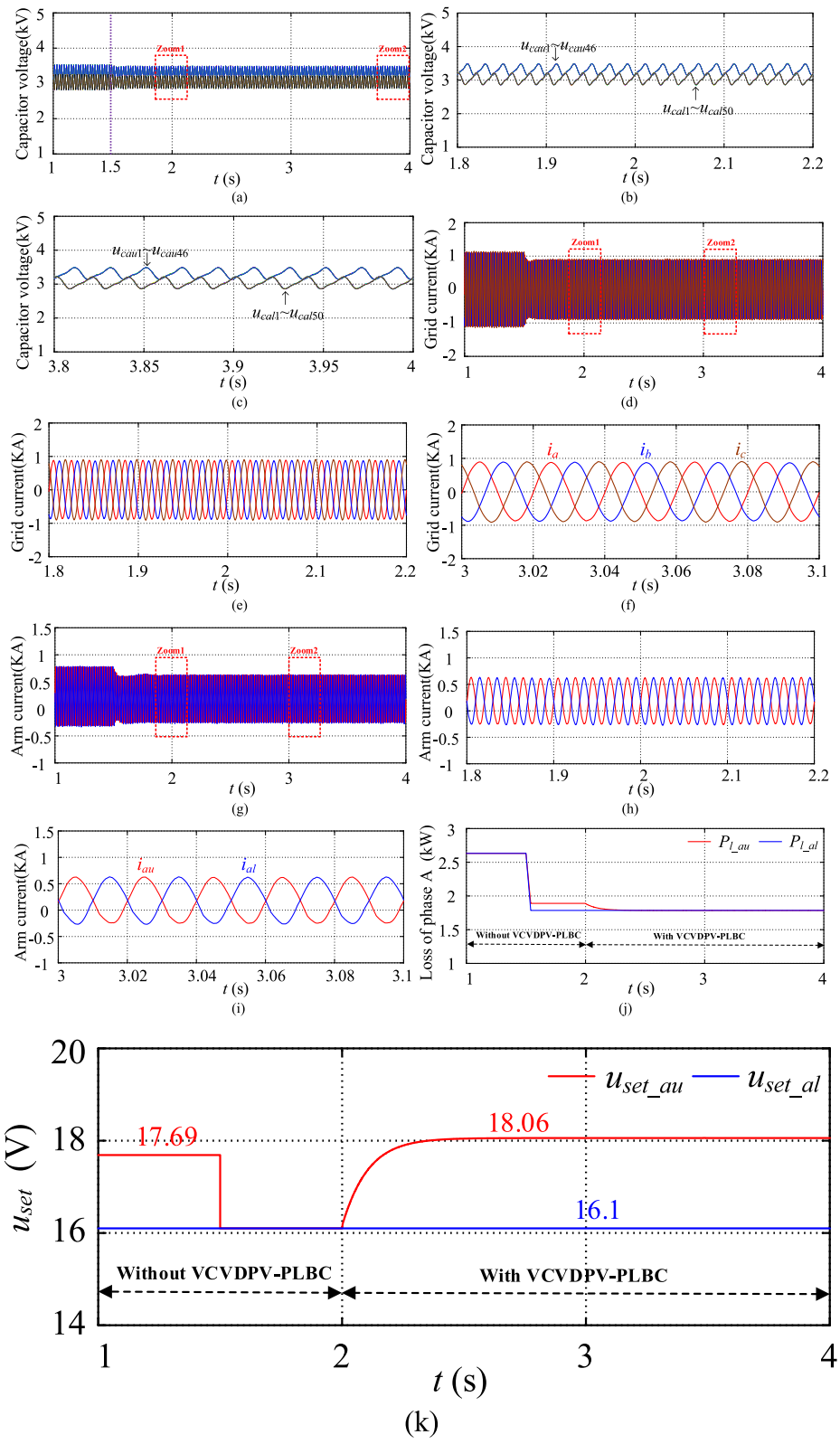


Fig. 15. (a) SM's capacitor voltages. (b) SM's capacitor voltages in Zoom1. (c) SM's capacitor voltages in Zoom2. (d) Grid currents. (e) Grid currents in Zoom1. (f) Grid currents in Zoom2. (g) Arm current. (h) Arm current in Zoom1. (i) Arm current in Zoom2. (j) Power loss in phase A. (k) Predefined value of capacitor voltage deviation.

current i_a including dynamic process in Zoom1. The $u_{cau1} \sim u_{cau2}$ are higher u_{cal1} and i_a has good output performance. Fig. 19(b) demonstrates the output voltage u_{cau}, the arm current i_{au} and i_{al}, the SM capacitor voltage $u_{cau1} \sim u_{cau2}$ and u_{cal1}, the output current i_a including steady state in Zoom2. The $u_{cau1} \sim u_{cau2}$ are higher u_{cal1} and i_a has good output performance. Obviously, the proposed VCVDPV-PLBC has few influences on the arm currents and the output current of the MMC in Fig. 19(b). Fig. 19(c) illustrates the power loss P_{l_au} and P_{l_al}, before 1.5 s,

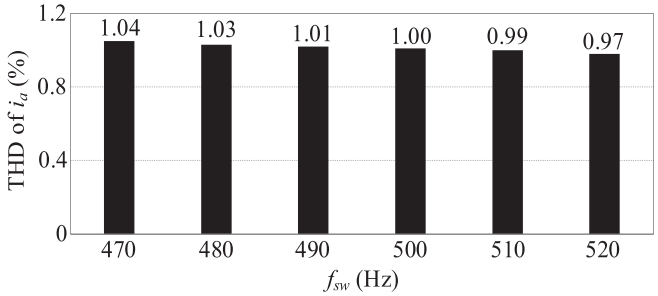


Fig. 16. THD of MMC's AC output current.

the power loss between the upper and lower arm is not equal., where the loss diversity is 0.113 W. Since 1.5 s, the approach which raised in this thesis is going to work, after a brief adjustment of the proposed method, the balance was regained for the P_{l_au} and P_{l_al}.

8. Conclusion

In this paper, the problem of unbalanced power consumption among arms caused by the failure of MMC's SM with more efficient spinning reserve is solved. By analysing the mathematical relationship between the capacitor voltage deviation predefined value, switching frequency and power loss, a VCVDPV-PLBC to improve the performance of MMC under SM fault is proposed. By changing the capacitor voltage deviation predefined value in MMC arm to change the switching frequency, the power loss among MMC arms can be balanced, thus improving the service life of MMC under SM malfunction. The effectiveness of the VCVDPV-PLBC is verified by building the relevant simulation and experiment. This article focuses on the issue of power loss imbalance between healthy and faulty arms caused by SM faults in arms, and studies the power loss balance control among different arms. Firstly, a capacitor voltage balance control method based on capacitor voltage deviation predefined value was introduced, and then the impact of SM faults on the capacitor voltage and switching frequency of SMs within the arm was analyzed. On this basis, the power losses of the MMC faulty arm and healthy arm were analysed under different SM fault ratios. SM faults result in imbalanced SM capacitor voltage and submodule switching frequency among arms, resulting in imbalanced power loss between faulty and healthy arms of the MMC, thereby reducing the reliability of the MMC. A VCVDPV-PLBC is proposed based on the relationship between the capacitor voltage deviation predefined value and power loss. By changing the capacitor voltage deviation predefined value to adjust the power loss, the power loss between the faulty arm and the healthy arm is balanced, thereby improving the reliability of the MMC when SM faults. At the same time, the impact of this method on the AC side power quality under different switching frequencies under submodule faults was analysed, and the influence of the VCVDPV-PLBC on the lifespan of power devices in the MMC was analyzed. Finally, the

Table 5
Experimental MMC's system arguments.

Arguments	Magnitude
SM's capacitance value (mF)	2.2
Arm's inductance value L_s (mH)	2
Modulation degree	0.85
SM numbers N_n	6
DC side voltage (V)	180
Load resistance R (Ω)	5

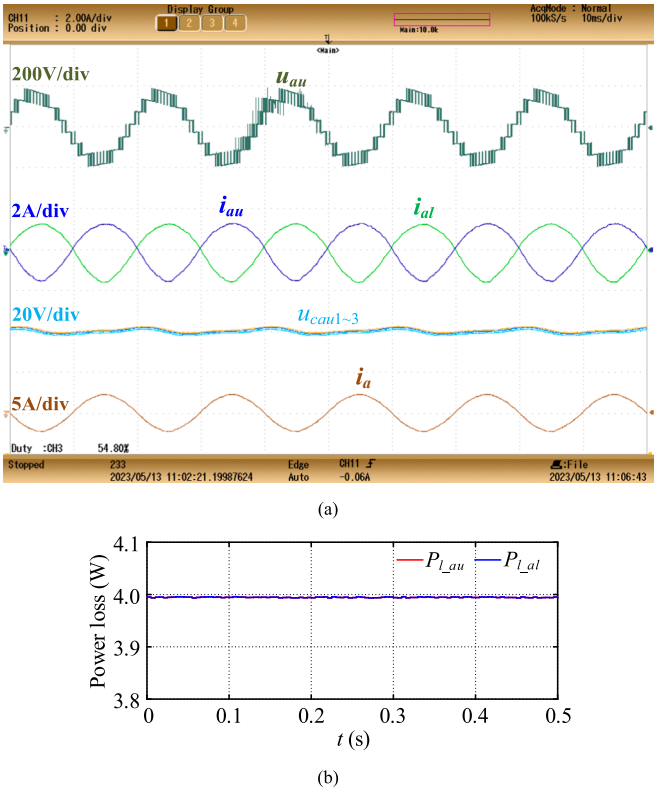


Fig. 18. The experimental results of the MMC under healthy situation. (a) u_{au}, i_{au}, i_{al}, $u_{cau1} \sim u_{cau3}$ and i_a. (b) P_{l_au} and P_{l_al}.

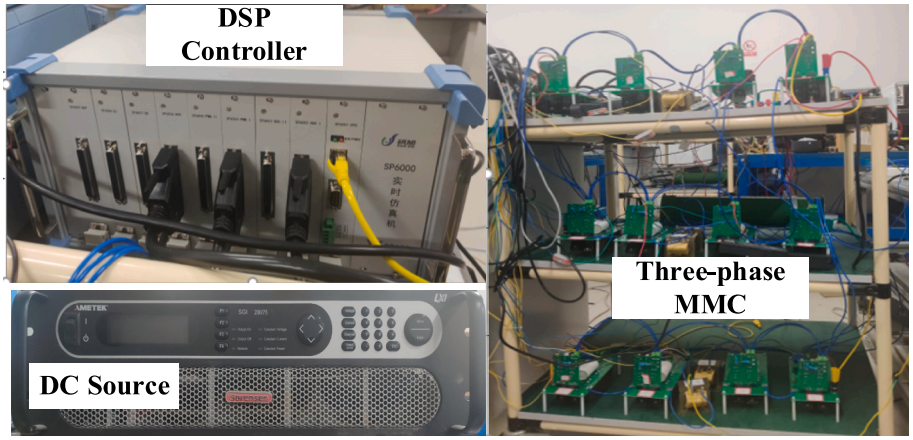
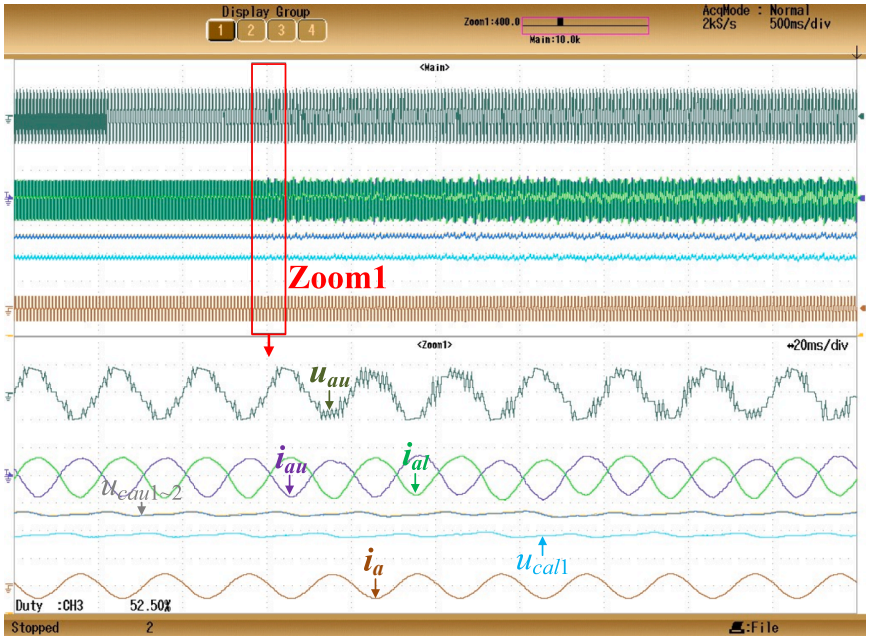
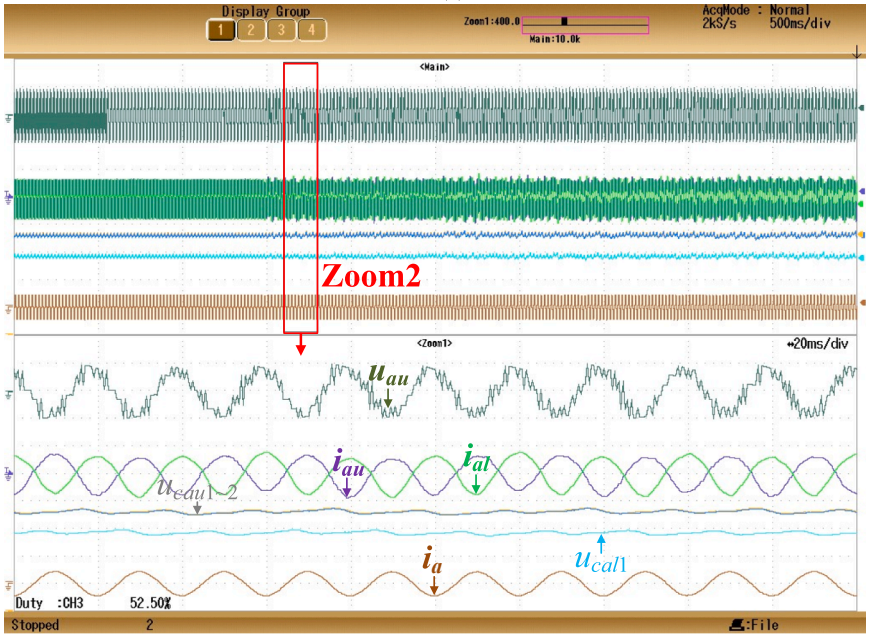


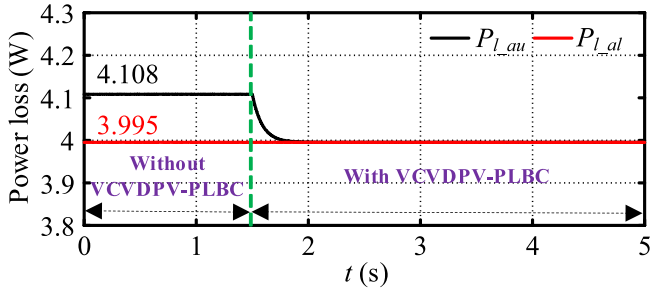
Fig. 17. Physical diagram of the MMC.



(a)



(b)



(c)

Fig. 19. Results obtained from the experiment of the MMC under SM faults situation. (a) u_{au}, i_{au}, i_{al}, $u_{cau1} \sim u_{cau2}$, u_{cal1} and i_a including dynamic process in Zoom1. (a) u_{au}, i_{au}, i_{al}, $u_{cau1} \sim u_{cau2}$, u_{cal1} and i_a including steady state in Zoom2. (c) P_{l_au} and P_{l_al}.

effectiveness of this control was demonstrated through simulation and experimental results.

CRediT authorship contribution statement

Jifeng Zhao: Conceptualization, Methodology, Writing – original draft. **Chaoyue Gu:** . **Sitong Wu:** Writing – original draft, Writing – review & editing, Investigation, Visualization. **Wenfeng Liu:** Visualization. **Yang Lei:** Investigation. **Jun Hang:** Writing – review & editing, Supervision. **Shichuan Ding:** Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

Acknowledgements

This work was supported by the Open Foundation of State Key Laboratory of High-end Compressor and System Technology (Project No. SKL-YSJ202209), the National Natural Science Foundation of China (Project No. 52377034, 52177027), the Key Project of Excellent Young Talents in University of Anhui Province (Project No. gxyqZD2021090) , the Excellent Youth Project of Natural Science Foundation of Anhui Province (Project No. 2108085Y18) and the Anhui Province Major Science and Technology Project (Project No. 202203c08020010).

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